Total No. of Questions : 6]

SEAT No. :

P46

[Total No. of Pages : 2

APR - 18/TE/Insem. - 48
T.E. (IT)
OPERATING SYSTEM
Information Technology
(2012 Course) (Semester - II) (314451)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:
1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.*
2) *Neat diagrams must be drawn wherever necessary.*
3) *Figures to the right side indicate full marks.*
4) *Assume suitable data, if necessary.*

Q1) a) Explain the concept of Virtual Machine with its implementation and benefits. **[6]**

b) List and explain operating system functions. **[4]**

OR

Q2) a) Explain the following operating systems **[6]**

i) Real-time systems.

ii) Distributed systems.

b) Explain traditional UNIX kernel architecture. **[4]**

Q3) a) Consider the following set of processes with the length of the CPU burst time given in milliseconds. **[6]**

Process	Arrival Time	Burst Time	Priority
P1	1	9	3
P2	0	3	1
P3	2	4	2
P4	4	3	3
P5	3	6	4

TQ = 1

Draw the Gantt charts illustrating the execution of these processes using RR. Calculate turnaround time and waiting time for each process.

b) Draw and explain the process state transition diagram. **[4]**

OR

Q4) a) Describe in detail the differences between short-term, medium-term and long-term schedulers with the help of a neat diagram. **[6]**

b) Write a note on user level and kernel level threads. **[4]**

Q5) a) Write a safety algorithm in the deadlock problem consider following snap shot of system at time T0. **[6]**

	Allocation			Max			Available		
	A	B	C	A	B	C	A	B	C
							2	3	0
P0	0	1	0	7	5	3			
P1	3	0	2	3	2	2			
P2	3	0	2	9	0	2			
P3	2	1	2	2	2	2			
P4	0	0	2	4	3	3			

Calculate the need matrix and state whether the system is safe. If yes, justify with stepwise calculations. Give safe sequence.

b) Explain different types of Semaphore. **[4]**

OR

Q6) a) Explain Producer Consumer Problem and write a solution using semaphore. **[6]**

b) What are the four conditions for deadlock to occur? **[4]**

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